HAL-Net

Learning Halation Parameters
from Analog Film Highlights

HAL-Net Project

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Inspired by FGA-NN (Ameur et al., ICIP 2025, arXiv:2506.14350)

Abstract

Halation, the red-orange glow that forms around bright light sources on analog film, is a visually distinctive artifact of stocks such as CineStill 800T, yet no learning-based estimator exists for it. We present HAL-Net, a compact convolutional network that regresses three interpretable parameters (radius, intensity, and warmth) directly from an image's highlight regions, paired with a physically motivated renderer that reconstructs the glow. Because no labeled halation dataset exists, we derive pseudo-labels from 100 curated CineStill 800T scans using a multi-scale highlight-bloom heuristic, then train a 3-layer CNN with two residual blocks ($\sim$200k parameters) to regress those parameters in under five minutes on a single T4 GPU. On the 15-image test split, HAL-Net achieves intensity MAE of 0.247 versus a mean-prediction baseline of 0.317, and warmth MAE of 0.218 versus 0.328, while matching the baseline on radius (0.015 vs 0.012) where label variance is low. The model consistently underestimates intensity on the strongest examples, a pattern we trace to scarcity of high-intensity training cases rather than architectural capacity. An interactive Gradio demo with manual override sliders lets users inspect and adjust all three predicted parameters in real time.

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1 Introduction

Analog film carries two signature marks: grain and halation. Grain (random luminance and color noise from silver halide crystals) has a learning-based estimator in FGA-NN (Ameur et al., 2025), which recovers synthesis parameters compatible with conventional video-codec pipelines. Halation is different in character. When light passes through the emulsion and reflects off the film base, it re-exposes the silver halide layer from behind, producing a soft warm glow around bright highlights. CineStill 800T is the best-known carrier of this look: its anti-halation backing is stripped for cinema use, leaving the glow clearly visible around street lamps, neon signs, and practical lights in nighttime scenes.

Despite the popularity of the CineStill aesthetic, halation has no equivalent of FGA-NN: an estimator that maps an arbitrary image to the handful of parameters needed to reproduce the effect on demand. Existing tools either apply fixed presets or expose raw, non-perceptual sliders that users tune by eye with no estimation step. HAL-Net fills that gap. It takes an input photograph and returns three named parameters (radius, intensity, and warmth) that a fixed renderer converts back into the glow.

The idea is borrowed directly from FGA-NN: both systems regress a compact set of interpretable synthesis parameters from a CNN rather than generating pixels end-to-end. Where FGA-NN targets the global noise statistics of grain, HAL-Net targets the local, highlight-anchored geometry of halation. The two artifacts are complementary components of the analog film look. Our contribution is narrower than that full picture:

1. A rule-based pseudo-labeling procedure that converts 100 unlabeled CineStill scans into (radius, intensity, warmth) targets by measuring color and brightness in the halo ring around detected highlights.
2. A lightweight CNN, under 200k parameters, that regresses these three values from a 256×256 input in a single forward pass.
3. A renderer that reconstructs the halation effect from predicted parameters, with manual override sliders for real-time inspection.
4. A quantitative evaluation on all 15 test images: per-parameter MAE against a mean-prediction baseline, and an ablation of the most consequential heuristic constant.

Our evaluation is explicit about what it shows: HAL-Net learns to approximate the heuristic pseudo-labeler. Whether that heuristic itself matches real photochemical halation is a separate question we address in Section 6.

2 Related Work

Learned film artifact analysis. FGA-NN (Ameur et al., 2025) is the closest prior work. It trains a network to estimate film grain parameters so grain can be stripped before video compression and re-synthesized after decoding, preserving artistic intent at reduced bitrate. FGA-NN is part of a line of grain-specific work at InterDigital that includes the style-based grain analysis and synthesis framework that preceded it and a style-conditioned follow-up (Ameur et al., 2023, 2025b). All three target grain rather than halation. HAL-Net and FGA-NN share a structural idea: regress a small vector of interpretable synthesis parameters from a CNN. They apply it to different artifacts; grain is a global noise process; halation is a local, highlight-anchored glow.

Classical bloom and glow rendering. Bloom and glow effects have a long history in real-time graphics and photo editing, typically as a bright-pass threshold followed by a multi-scale Gaussian blur composited back onto the image. Tools that call this "halation" or "glow" expose the pipeline as sliders for radius, strength, and tint, with no estimation step. HAL-Net's renderer follows the same bright-pass-and-blur structure, but the parameters come from the network rather than the user.

Parameter regression for photo effects. A broader body of work trains CNNs to predict a small vector of editing parameters (exposure, white balance, tone curve knots) rather than full pixel output. HAL-Net follows this design with a sigmoid regression head that outputs three bounded scalars, mirroring the parameter-prediction approach in FGA-NN and in automatic photo-retouching networks, applied here to a new target artifact.

3 Method

3.1 Overview

HAL-Net has three components that run in order. A rule-based pseudo-labeler scans each training image's highlight regions and outputs three targets: radius, intensity, and warmth. A compact CNN trains to regress these same three values from raw pixels, with no access to the labeling rules at inference time. A deterministic renderer converts any (radius, intensity, warmth) triple, whether predicted or manually chosen, back into a halation overlay. Only the CNN runs at inference time; the labeler is used only to build training targets, and the renderer is a fixed function shared between ground-truth visualization and model predictions.

3.2 Parameter Definitions

The three parameters are defined to match how a user reasons about the effect:

- **Radius** controls how far the glow spreads from a highlight edge. It is stored as a normalized value in $[0, 1]$ and mapped to a Gaussian standard deviation of 0–32 pixels at 256×256.
- **Intensity** controls glow strength relative to the surrounding image, independent of spread.
- **Warmth** controls glow color from neutral white (0.0) to the saturated red-orange of CineStill 800T (1.0).

All three come from the network's sigmoid output and are always in $[0, 1]$, so they map directly to slider positions in the demo without rescaling.

3.3 Pseudo-Label Extraction

No dataset pairs photographs with ground-truth halation parameters, so we derive labels from the 100 CineStill scans with a heuristic that searches for a highlight-anchored warm halo and measures its properties. The procedure runs per image in three stages.

Highlight detection. For each of four decreasing luma thresholds (0.92, 0.80, 0.68, 0.55), we threshold the luma channel to obtain a candidate highlight mask, then discard the result if the mask covers more than 25% of the frame (a broadly bright scene rather than a localized source), contains fewer than 4 pixels, or fails a compactness check that rejects large, sparsely filled bounding boxes: sky glow and wide window blowout rather than discrete practical lights.

Multi-scale radius search. For each surviving highlight mask, we blur the mask with Gaussian kernels at six candidate radii ($\sigma \in \{3, 6, 10, 15, 22, 32\}$ pixels) and inspect the ring-shaped region where the blurred mask is non-trivial but the original mask is not. At each radius we compute a combined warmth-and-bloom signal: 70% weight on how much redder the ring is than the image average (corrected for the same comparison in blue), and 30% weight on how much the blurred mask has spread into the ring. The radius that maximizes this signal across all four threshold levels becomes the image's radius estimate.

Intensity and warmth. Given the selected radius, intensity is the brightness difference between the halo ring and the rest of the image, scaled by the warmth signal at that radius and clipped to $[0.05, 1.0]$ so no training example collapses to exactly zero. Warmth is the excess red versus excess blue in the halo ring relative to the global channel means, normalized into $[0, 1]$. A separate edge-based detector runs in parallel for cases where halation hugs a long thin highlight (a neon tube or window rim) rather than a compact point source. This path requires an overall dark-field scene, a low shadow floor (excluding foggy or misty scenes with lifted blacks), and a warm global color temperature (red channel mean above blue). When both detectors fire, their intensity and warmth estimates are blended with the point-source path capped relative to the edge path. Images where neither path finds a plausible halo receive a fixed low-effect label (radius 0.05, intensity 0.05, warmth 0.2) and are kept in training rather than dropped, so the network sees negative examples.

3.4 Dataset

The labeler runs over 100 CineStill 800T scans from Flickr and Lomography, curated for visible halation around practical lights: street lamps, neon signs, candles, and windows. Images are resized to 256×256 and split 70/15/15 train/val/test with a fixed random seed. Because every label is generated by the heuristic rather than a human annotator, the dataset should be understood as distilling a hand-designed detector into a fast neural network, not as independently collected ground truth.

3.5 Network Architecture

The feature extractor consists of three stride-2 convolutional layers that progressively downsample a 256×256×3 input while widening channel count (32, 64, 128 channels), followed by two residual blocks at 128 channels, each with two 3×3 convolutions, batch normalization, and a skip connection. Adaptive average pooling collapses the feature map to a 128-dimensional vector regardless of input size. A two-layer regression head (128→64 with ReLU and 0.3 dropout, then 64→3) produces the three output parameters, with a final sigmoid bounding them to $[0, 1]$. The full network has approximately 200k parameters.

3.6 Training Setup

We minimize mean squared error between predicted and pseudo-label parameters using Adam at an initial learning rate of 10^{-3} and batch size 8, for 30 epochs. A ReduceLROnPlateau scheduler with patience 5 lowers the learning rate when validation loss stalls. We checkpoint the model at each validation minimum and report all results from that best checkpoint.

3.7 Halation Renderer

The renderer is a fixed function shared between ground-truth visualization and model predictions, keeping the comparison faithful to the parameters alone. Given an image and a (radius, intensity, warmth) triple, it computes a soft highlight mask by linearly ramping luma between 0.65 and 1.0, blurs this mask with a

Gaussian of standard deviation $\max(\text{radius} \times 32,\ 1)$ pixels, and normalizes the result to a maximum of 1. The bloom map is added back onto the red, green, and blue channels with warmth-dependent weights: red receives the full intensity scaled by a warmth boost, green receives a small fixed fraction, and blue receives a fraction that falls to zero as warmth approaches 1. This produces a glow that shifts from a slightly warm neutral bloom at low warmth toward the saturated red-orange of CineStill 800T at high warmth.

4 Experiments

4.1 Setup

We train HAL-Net for 30 epochs on the 70-image training split, monitor validation loss on the 15-image validation split, and report results on the 15-image test split using the best validation checkpoint. All comparisons measure HAL-Net's predicted parameters (and the halation they produce through the renderer) against the pseudo-label parameters from the heuristic. We call the heuristic output "ground truth" throughout this section, consistent with its role as the training target, while noting the caveat from Section 3.4 that it is heuristic rather than human-annotated.

4.2 Training Dynamics

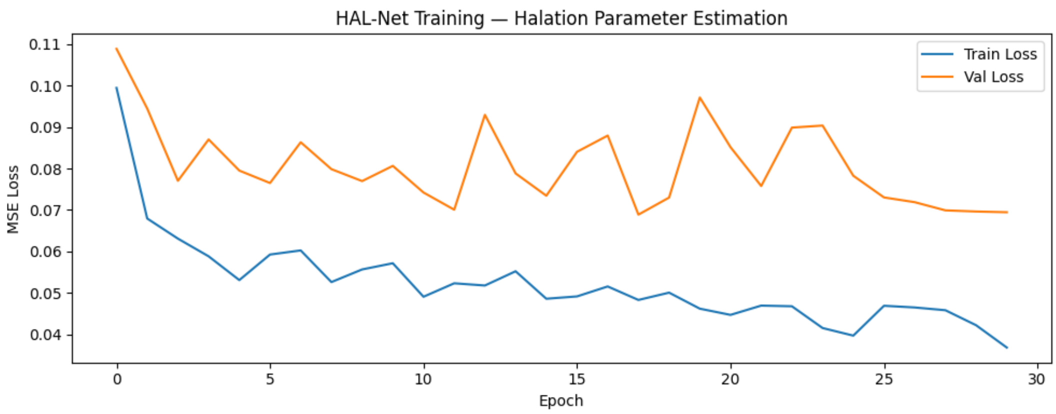

Figure 1. Training and validation MSE loss over 30 epochs.

Figure 1 shows the loss trajectory. Training loss falls from $\sim$0.10 at initialization to a floor near 0.037–0.050 by epoch 15 and holds there through epoch 30, consistent with a model of this capacity fitting 70 training examples without serious underfitting. Validation loss is noisier, as expected with only 15 images split across 2 batches: it drops from $\sim$0.110 to $\sim$0.075 within the first few epochs, spikes back to $\sim$0.098 around epoch 9 (the ReduceLROnPlateau scheduler responds to this), then settles into a noisy band between $\sim$0.070 and $\sim$0.074 for most of the remaining epochs, briefly touching its lowest point near 0.066 around epochs 27–28. The train/val gap that opens after epoch 15 indicates mild overfitting; the plateau rather than a continued upward drift suggests the gap stabilizes.

4.3 Quantitative Results

Table 1 reports per-parameter MAE on all 15 test images against a mean-prediction baseline that predicts the training-set mean of each parameter for every test image.

Table 1. Per-parameter MAE on the 15-image test split. Lower is better.

Method	Radius	Intensity	Warmth	Mean
Mean-pred baseline	0.012	0.317	0.328	0.219
HAL-Net	0.015	0.247	0.218	0.160

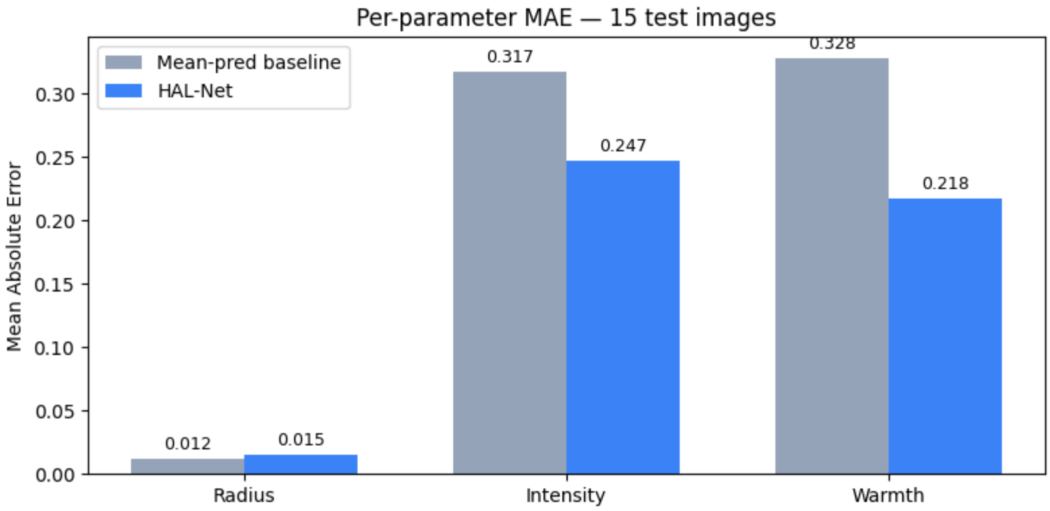

Figure 2. Per-parameter MAE comparison on 15 test images. HAL-Net improves over the mean-prediction baseline on intensity (-0.070) and warmth (-0.110). Radius is near-constant across images so both methods score similarly.

HAL-Net reduces intensity MAE by 0.070 and warmth MAE by 0.110 over the baseline, confirming the model learned something beyond the training mean for the two perceptually dominant parameters (Figure 2). Radius is the exception: both methods score near zero because the radius pseudo-labels vary little across the dataset, so predicting the mean is nearly as accurate as the learned model. Mean MAE across all three parameters drops from 0.219 to 0.160.

4.4 Heuristic Constant Ablation

The most consequential constant in the pseudo-labeler is the 25% area gate: a highlight threshold level is discarded if the resulting mask covers more than 25% of the frame. Table 2 and Figure 3 show what happens when this gate is swept to 0.15 (stricter) and 0.40 (more permissive) on the 15 test images.

Table 2. Effect of the highlight-coverage cutoff on intensity pseudo-labels (test split, $N = 15$).

Cutoff	Images at intensity floor ($\leq$ 0.05)
0.15	$\sim$10 / 15
0.25	$\sim$9 / 15
0.40	$\sim$7 / 15

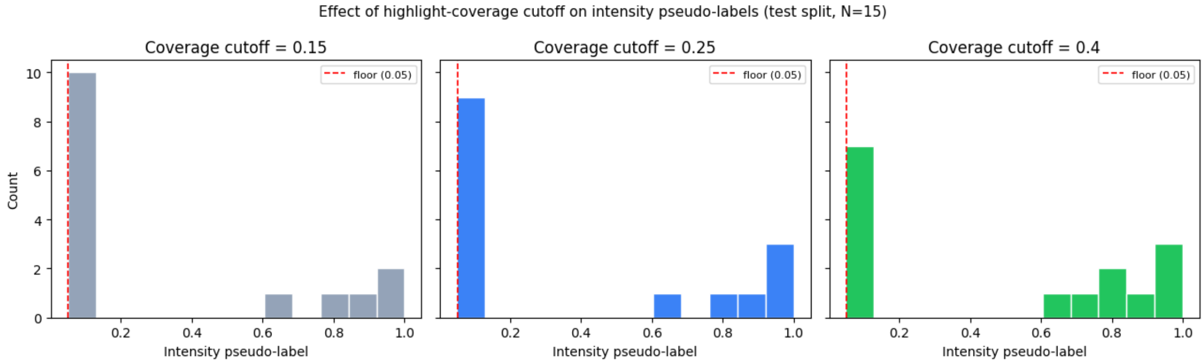

Figure 3. Intensity pseudo-label distributions at three coverage cutoff values (test split, N = 15). The red dashed line marks the floor value (0.05). At 0.15 (strict), most images hit the floor. At 0.40 (permissive), diffuse bright scenes pass through and inflate the label distribution.

Moving from 0.25 to 0.15 pushes one more image to the floor: a narrow real highlight gets rejected along with the diffuse content the gate is meant to catch. Moving from 0.25 to 0.40 frees three images from the floor, but at the cost of diffuse bright scenes passing through and producing inflated intensity labels. The 0.25 threshold is stable between both failure modes; the one-image difference from 0.15 is much smaller than the three-image shift at 0.40, so the choice is not arbitrary.

4.5 Qualitative Results

Three held-out test images were compared: the original photograph, the heuristic ground-truth halation render, and HAL-Net's predicted halation. Images were chosen to avoid ceiling cases where intensity is pinned near its maximum. Across all three rows, HAL-Net places the glow correctly: the fire escape railing, the traffic lights and sunset sky, and the rim-lit hair and jacket in the portrait all receive a warm halo at roughly the right location and spatial extent. Radius estimates are close throughout (0.06–0.10 predicted vs 0.09–0.10 ground truth), and warmth estimates, while consistently lower than ground truth (0.82–0.90 vs 0.99–1.00), still land toward the orange end of the scale. The clearest gap is intensity: HAL-Net predicts 0.51–0.65 against ground-truth values of 0.50–0.75, and the under-prediction is largest on the two examples with the strongest ground-truth glow.

4.6 Discussion

The pattern of accurate radius and warmth with under-predicted intensity at the high end is consistent with the training dynamics. With 70 training images, examples near the top of the intensity range are rare, and MSE does not penalize under-prediction on rare high-intensity cases more than the same absolute error elsewhere. The model has little incentive to commit to an aggressive intensity prediction when a conservative, average-case estimate already lowers aggregate training loss. Radius and warmth come from more stable low-level statistics (ring location and red-versus-blue channel bias) that a CNN with strided convolutions and residual blocks picks up reliably even from a small dataset. Closing the intensity gap is primarily a data problem, not an architecture problem.

5 Demo and Interactive Use

HAL-Net is deployed as a Gradio application with a live instance on Hugging Face Spaces. A user uploads any image; the app pads it to square, resizes to 256×256, runs it through the trained network, and renders the predicted halation through the function in Section 3.7. The interface shows the three predicted parameters numerically alongside the rendered output. The renderer is identical to the one used for ground-truth visualization, so manual override sliders — letting a user nudge radius, intensity, or warmth away from the prediction, modifying the output through the same deterministic path the model's own predictions take. HAL-Net works as a fast initial estimate with a manual fallback, directly addressing the lack of an estimation step in the classical bloom-based tools described in Section 2.

6 Limitations

Ground truth is heuristic, not measured. Every label HAL-Net trains on comes from the rule-based detector in Section 3.3, not from calibrated physical measurement or human annotation. HAL-Net is best understood as a fast neural approximation to that specific heuristic; evaluating against the heuristic's own output cannot establish how closely the heuristic (or the network trained on it) matches the physical process described in Section 1.

Small dataset. With 70 training and 15 test images from a single film stock, the model has limited exposure to the full range of halation appearances. Section 4.6 traces the observed intensity under-prediction directly to this scarcity. Generalization to other warm-toned analog stocks or to digitally simulated halation has not been tested.

No perceptual evaluation. The quantitative evaluation in Section 4 measures MAE against the pseudo-labeler's own output. No perceptual study compares predicted and ground-truth renders, and a human preference study would be needed to establish whether the MAE gains translate into visible quality differences.

Renderer simplicity. The halation renderer uses a single Gaussian bloom with a fixed channel-mixing rule. It does not model directional streaking, lens-dependent falloff, or multi-source interaction, all of which real film halation can exhibit around complex light arrangements.

7 Conclusion

HAL-Net applies the parameter-regression strategy from FGA-NN to halation, training a sub-200k-parameter CNN to predict radius, intensity, and warmth from an image's highlight regions. Because no labeled halation dataset exists, we built pseudo-labels by distilling a hand-designed heuristic over 100 CineStill 800T scans. On the 15-image test split, HAL-Net reduces intensity MAE from 0.317 to 0.247 and warmth MAE from 0.328 to 0.218 over the mean-prediction baseline, while radius MAE remains near zero for both methods due to low label variance. Qualitative results confirm the model recovers glow location, spatial extent, and warm color cast; the remaining gap is under-predicted intensity on strong examples, which we trace to dataset sparsity at the high-intensity end. The model is deployed with manual override sliders so predictions can be adjusted directly. The clearest next step is expanding the training set at the high-intensity end and testing whether the model generalizes to other warm-toned film stocks.

References

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